# Half-wave rectifier: filter for mean output and ripple

The first config-driven GA run that is nonlinear and transient — value genome, milestone M5 (with the M4 diode), CLAUDE.md §4

kronspice simulation report · 2026-11-07 · population 30, seed 11

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## Summary

A 10 V peak, 1 kHz sine drives a diode into a filter capacitor and a load resistor. A genetic algorithm chooses the capacitor, the load resistor and the diode's series resistance so that, once the start-up transient has settled, the output averages 9 V with no more than 0.2 V of peak-to-peak ripple. It is the first config-driven example that needs the nonlinear Newton solve and a transient analysis for every candidate.

- **Result (seed 11):** C1 = 820 µF, R1 = 270 Ω, diode RS = 0.119 Ω, reached after generation 13 (394 evaluations).
- **Mean output 9.007 V** over 4–8 ms, 0.08 % from the 9 V target, re-simulated from the written *best.cir*.
- **Ripple 0.154 V** peak to peak, under the 0.2 V limit.
- **Why it works:** R1·C1 = 221 ms against the 1 ms period holds the ripple down; the mean is then set by the crest voltage, 10 V less the diode drop at the charging peak.
- **A .model parameter evolves like a part value:** the diode's RS sets the charging current's peak and the drop at the crest.

## Background

CLAUDE.md §4 describes the value genome (fixed topology, component values on a log scale, optional E-series snapping); M5 adds genes that target a *.model* parameter. This example exercises that together with the M4 diode and its Newton fallback chain (doc/M4.md). It is a standard engineering exercise; no hypothesis is involved. The other config-driven M5 examples are *rc_lowpass_values* and *negative_resistance_gain*.

## Circuit, genes and objectives

Template netlist (*circuit.cir*), with the starting values:

```
Half-wave rectifier: evolve the filter capacitor and load for ripple and mean output
V1 in 0 SIN(0 10 1k)
D1 in out dr
R1 out 0 1k
C1 out 0 1u
.model dr D(IS=1e-14 N=1 RS=0.5)
.tran 10u 8m
.end
```

| Gene | Range | Series |
|---|---|---|
| C1 | 100 nF … 1 mF | E12 |
| R1 | 100 Ω … 100 kΩ | E24 |
| dr RS (a .model parameter) | 0.05 … 50 Ω | continuous (log scale) |

| Objective | Window | Target | Error |
|---|---|---|---|
| mean of V(out) | 4 ms – 8 ms | 9.0 V | \|mean − 9\| / 9 |
| peak-to-peak ripple of V(out) | 4 ms – 8 ms | ≤ 0.2 V | max(0, ripple − 0.2) / 0.2 |

Both objectives weigh 1; the ripple is a constraint, costing nothing once it is met. GA settings: population 30, up to 25 generations, seed 11, a 5 s deadline per evaluation and *target_fitness* 0.002.



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## Result

| Part / measure | Value |
|---|---|
| C1 | 820 µF |
| R1 | 270 Ω |
| dr RS | 0.119 Ω |
| Mean V(out), 4–8 ms | 9.007 V (0.08 % from target) |
| Ripple, 4–8 ms | 0.154 V (limit 0.2 V) |
| Time constant R1·C1 | 221 ms (period 1 ms) |
| Generations / evaluations | target reached after generation 13 / 394 |

doc/M5.md summarizes the run as "within 0.08 %, ripple met".

## Validation

- Every transient is checked by the energy ledger as it runs: a candidate whose books do not balance is penalized, never reported. Per doc/M5.md, each point's residual is judged against the whole run's peak power after the run, and gross imbalances ($> 10^{-3}$) stop a run immediately.
- The figures above are from re-simulating the written *best.cir*, not from the GA's own record.
- Exported template (*export/rectifier_filter_circuit_results.md*, 4 elements, 2 nodes): operating-point sheet only. Kron's torn solution is shown for linear circuits; this one contains a diode, so the sheet reports the MNA solution (the torn Newton solve, *kron.JacobianTearing*, is checked in kron's tests).
- Run outputs go to *runs/<timestamp>-rectifier_filter/* (*stats.csv*, *best/genNNNN.cir*, *best.cir*, *checkpoint.json*); *best.cir* opens in ngspice, LTspice and QSpice as it is.

## Limits

- The diode is the SPICE3 level-1 model with IS = $10^{-14}$ A and N = 1; only RS evolves.
- Mean and ripple are measured over the 4–8 ms window of an 8 ms run.
- The source is ideal; source impedance, diode recovery and capacitor ESR are not modelled.

## Reproducing the results

```
go run ./cmd/evolve -config examples/rectifier_filter/config.json   # about 20 s
KRONSPICE_EXPORT=1 go test ./internal/export                         # writes export/
```

See also doc/M5.md (the value-genome engine) and doc/M4.md (the diode and the Newton fallback chain).


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